Procesamiento de imágenes médicas

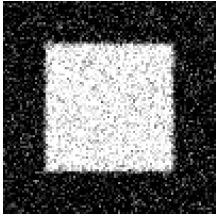
Ruido y degradación

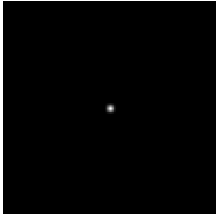
Alejandro Veloz

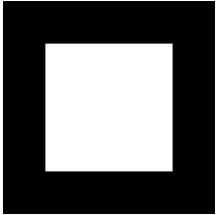
Modelo de degradación

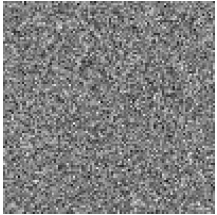
Un supuesto común para los mecanismos de adquisición de imágenes es un sistema LTI:

$$\underbrace{g(x,y)}_{\text{imagen adquirida (observada)}} = \underbrace{h(x,y)}_{\text{mecanismo de adquisición}} * \underbrace{f(x,y)}_{\text{imagen real (no observada)}} + \underbrace{\eta(x,y)}_{\text{ruido}}$$









Ruido

Ruido tipo Gaussiano

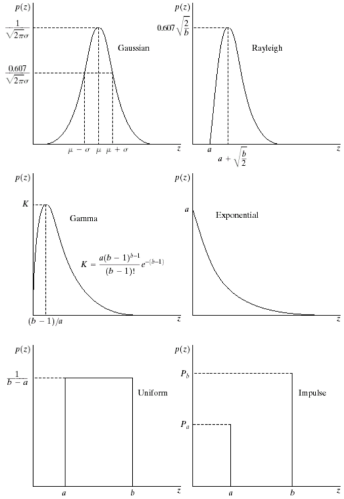
Ruido tipo Rayleigh

Ruido tipo Erlang (Gamma)

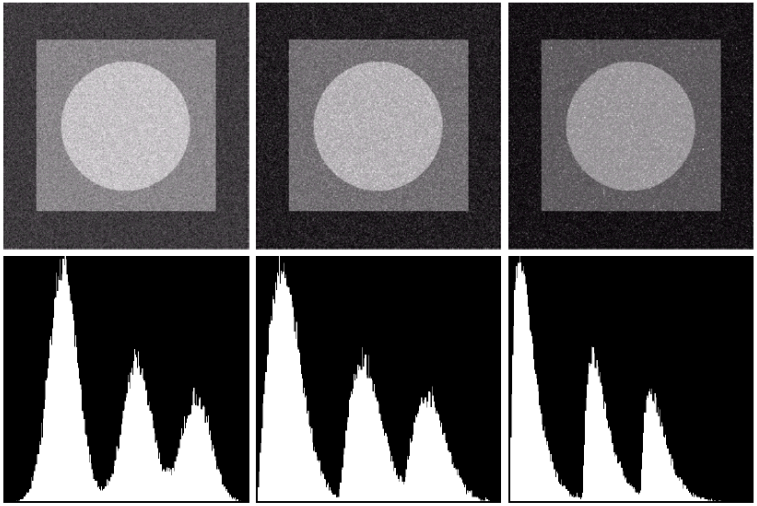
Ruido tipo Exponencial

Ruido tipo Uniforme

Ruido tipo impulsivo (salt-andpepper)

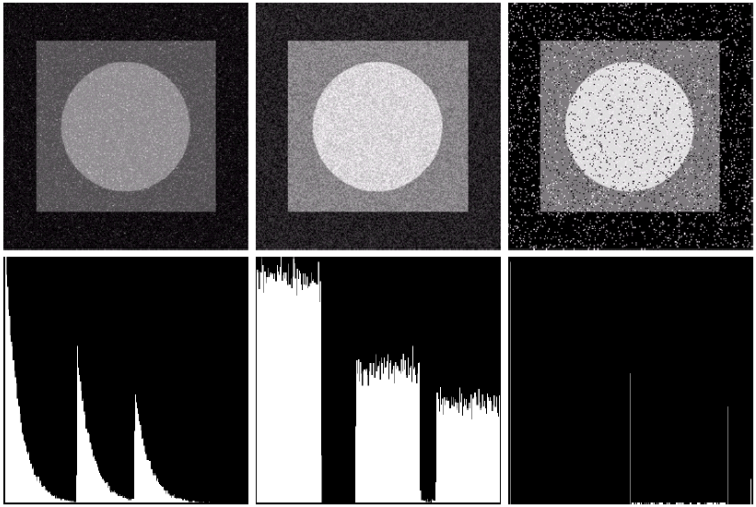


Ruido



Gaussian Rayleigh Gamma

Ruido

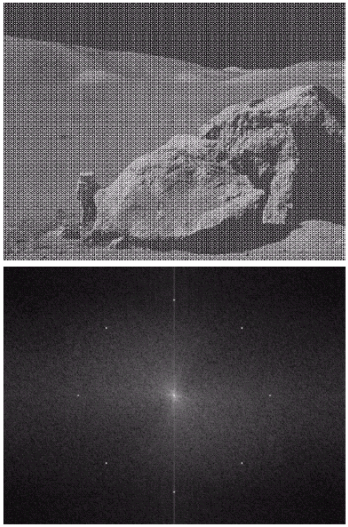


Exponential Uniform Salt & Pepper

Ruido

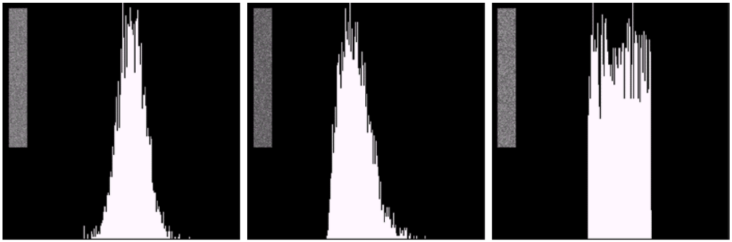
a
b

FIGURE 5.5 (a) Image corrupted by sinusoidal noise. (b) Spectrum (each pair of conjugate impulses corresponds to one sine wave). (Original image courtesy of NASA.)



Ruido

Es posible estimar los parámetros de las PDF a partir de un conjunto de “ventanas” de intensidad homogénea.



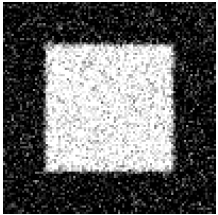
a b c

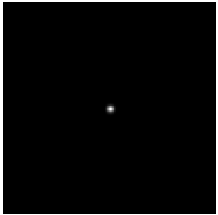
FIGURE 5.6 Histograms computed using small strips (shown as inserts) from (a) the Gaussian, (b) the Rayleigh, and (c) the uniform noisy images in Fig. 5.4.

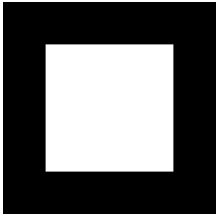
Estimación de $h(x, y)$

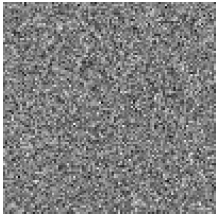
Para determinar $h(x, y)$ se debe tener en cuenta que sólo se conoce $g(x, y)$.

$$\underbrace{g(x,y)}_{\substack{\text{imagen adquirida}\\ \text{(observada)}}} = \underbrace{h(x,y)}_{?} * \underbrace{f(x,y)}_{?} + \underbrace{\eta(x,y)}_{\text{ruido}}$$





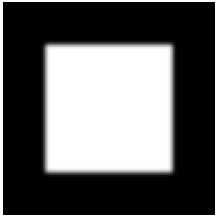


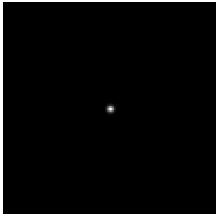


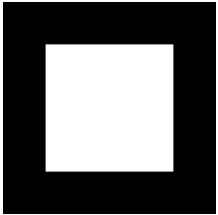
Estimación de $h(x, y)$

Para determinar $h(x, y)$ se debe tener en cuenta que sólo se conoce $g(x, y)$.

$$\underbrace{g_{\hat{\eta}}(x, y) = g(x, y) - \hat{\eta}(x, y)}_{\text{imagen filtrada}} \quad = \quad \underbrace{h(x, y)}_{?} \quad * \quad \underbrace{f(x, y)}_{?}$$







Estimación de $h(x, y)$

Para determinar $h(x, y)$ se debe tener en cuenta que sólo se conoce $g(x, y)$.

$$g_{\hat{\eta}}(x, y) = h(x, y) * f(x, y)$$

Optimizar alternadamente para $h(x, y)$ y para $f(x, y)$.

$$\hat{h}_{\mathrm{it+1}}(x, y) = \underset{h(x,y)}{\text{arg min}} \left\| g_{\hat{\eta}}(x, y) - h(x, y) * f_{\mathrm{it}}(x, y) \right\|_F^2 + \lambda \left\| h(x, y) \right\|_F^2$$

$$\text{s.t. } h(x, y) \in \mathbb{S}_{0,1}$$

$$\hat{f}_{\mathrm{it+1}}(x, y) = \underset{f(x,y)}{\text{arg min}} \left\| g_{\hat{\eta}}(x, y) - h_{\mathrm{it}}(x, y) * f(x, y) \right\|_F^2 + \lambda \left\| f(x, y) \right\|_F^2$$